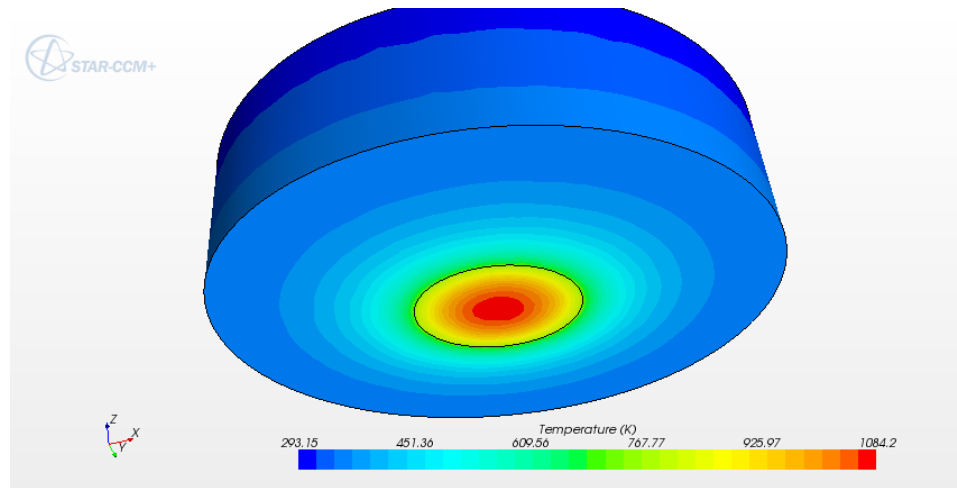


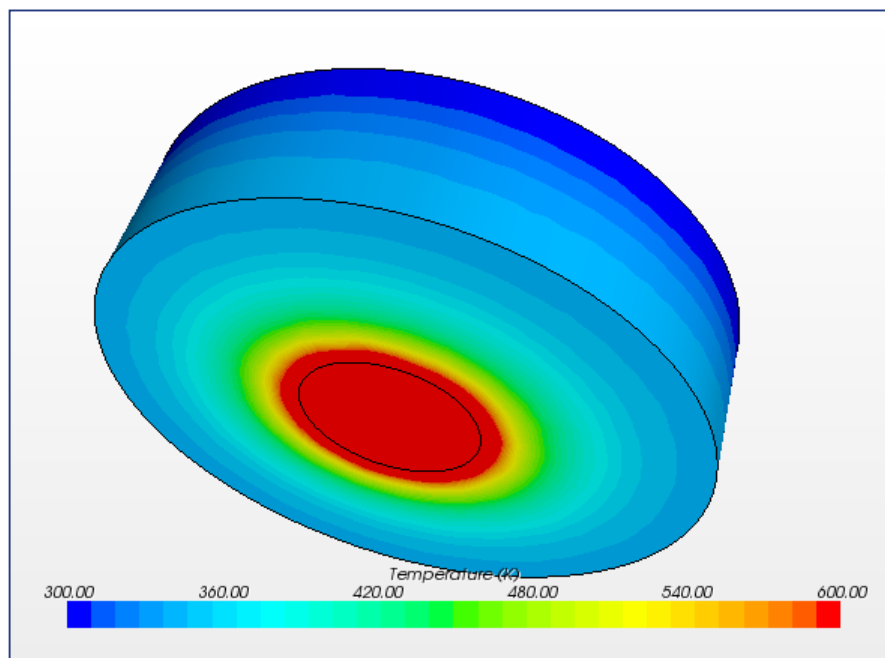
Heat Transfer Workshop (2014/11/14)



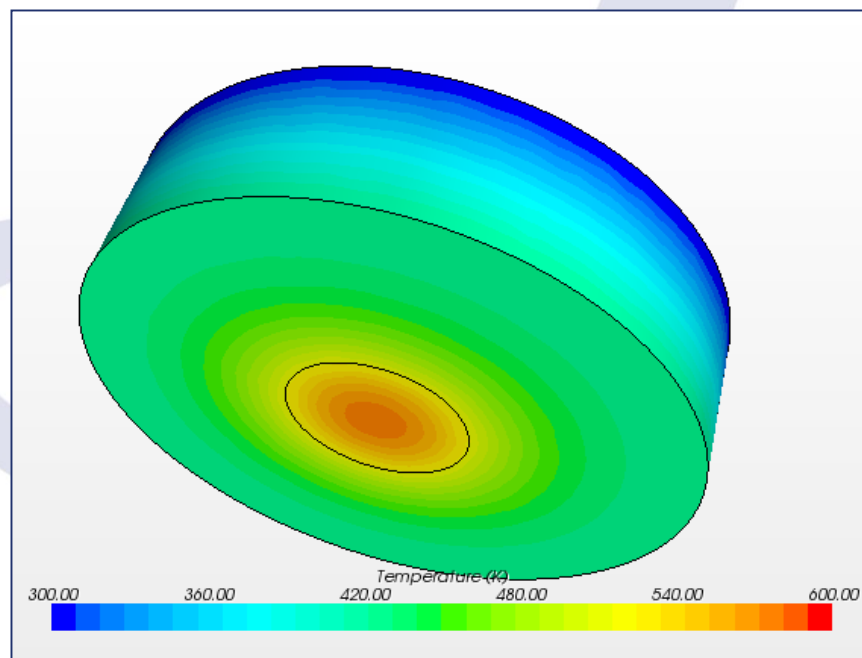
Workshop4 Shell模拟

Shell模拟

- Shell模拟
 - 在比较薄的固体Region中，面内的热传导比较重要的场合，可以使用这样的模拟方法。



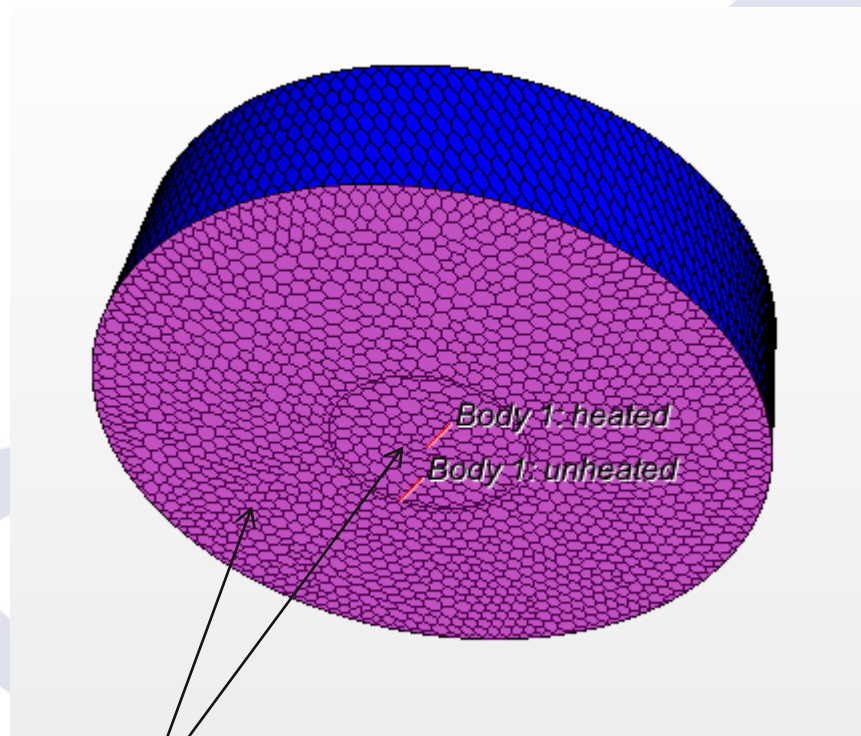
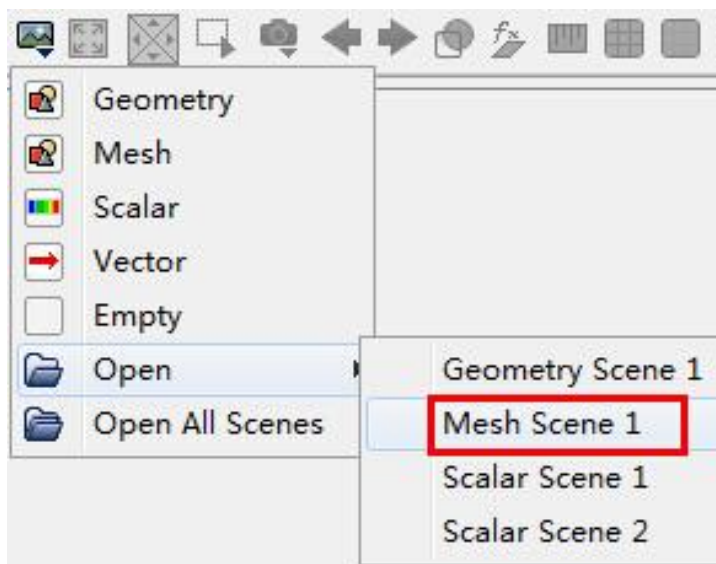
不考虑面内方向的传导



假定一个薄的固体层，考虑面内方向的热传导
(使用Shell模拟功能)

生成三维Shell物理模型

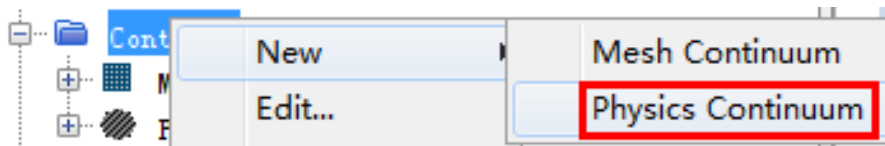
1. File > Load Simulation > ShellModel.sim
2. 打开已有的网格视图



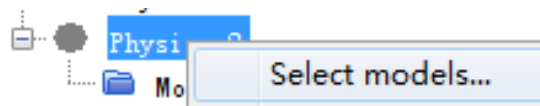
底面的壁面边界使用Shell来进行模拟。

生成三维Shell物理模型

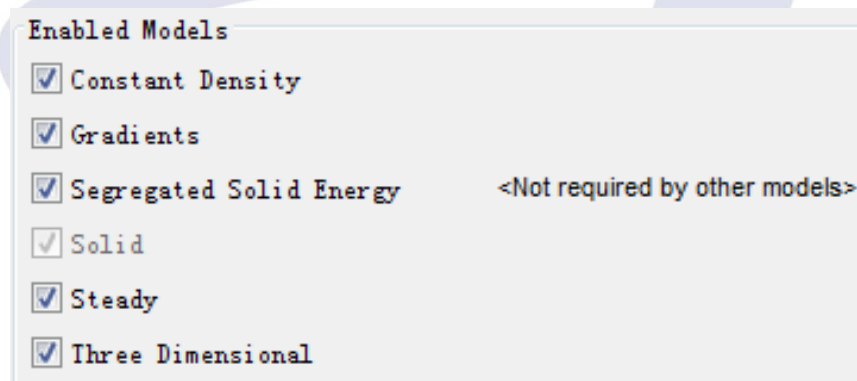
1. 右击[Continua]> [New] > [Physics Continuum]



2. 右击[Physics 2]、选择[Select models]

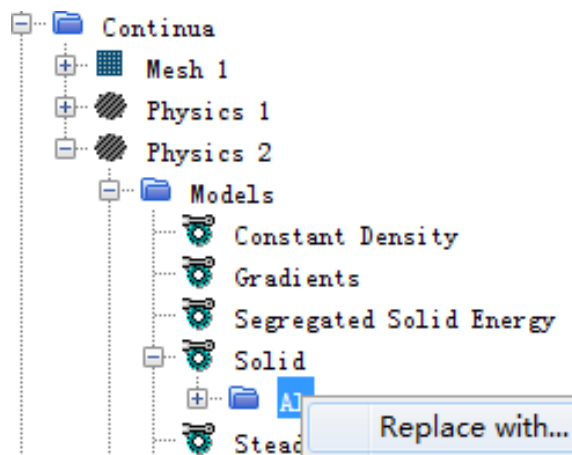


- Three Dimensional
- Solid
- Segregated Solid Energy
- Constant Density
- Steady

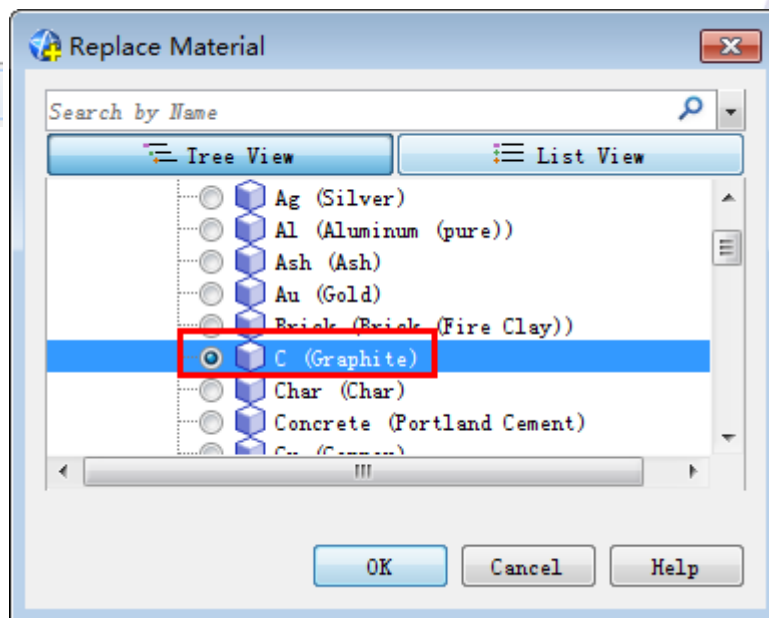


生成三维Shell物理模型

1. [Physics 2] > [Models] > [Solid] > [Replace with]

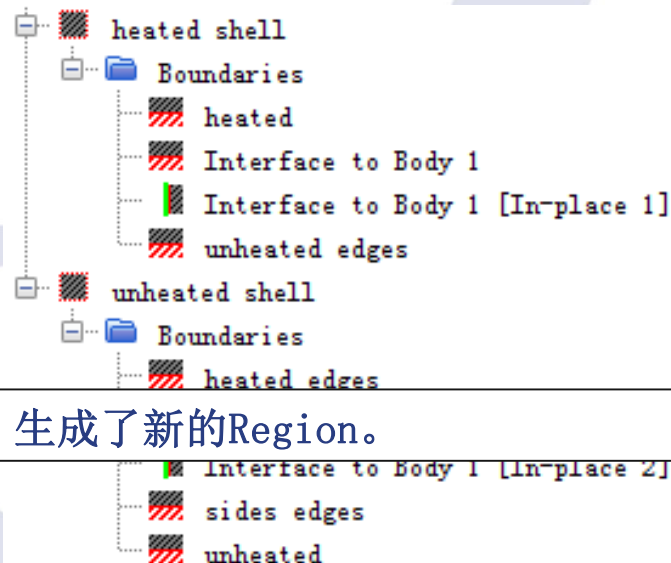
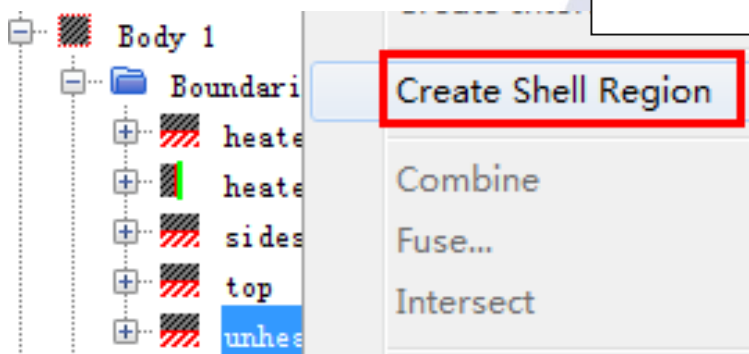
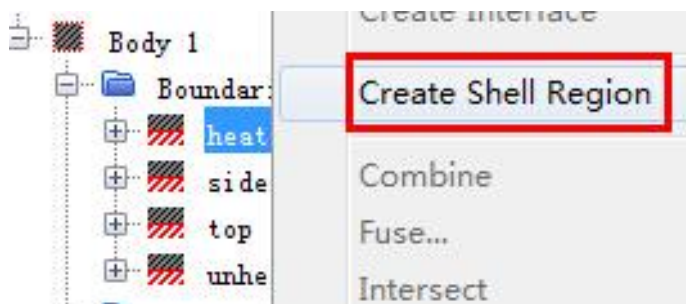


2. 在数据库中选择“C”。



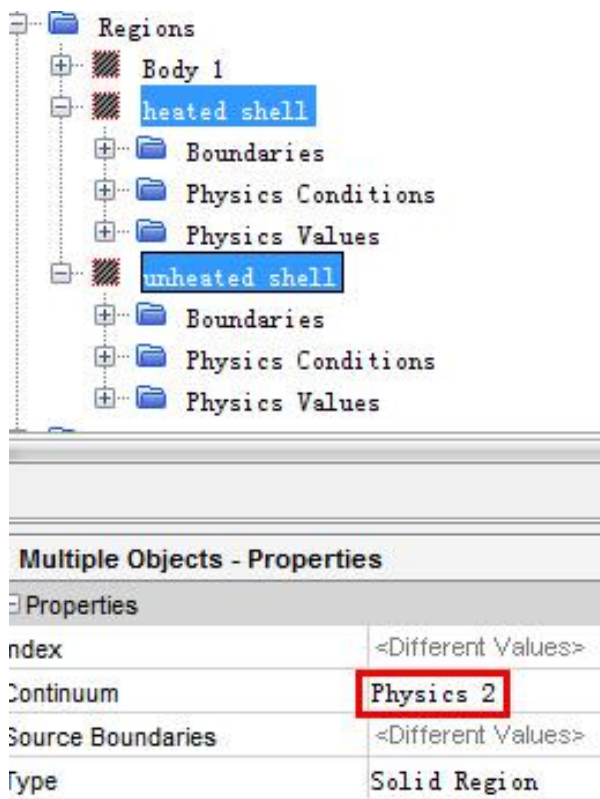
生成Shell Region

1. 右击[Boundaries] > [heated], 选择 [Create Shell Region]
2. 右击[Boundaries] > [unheated], 选择 [Create Shell Region]



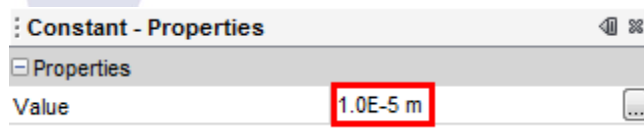
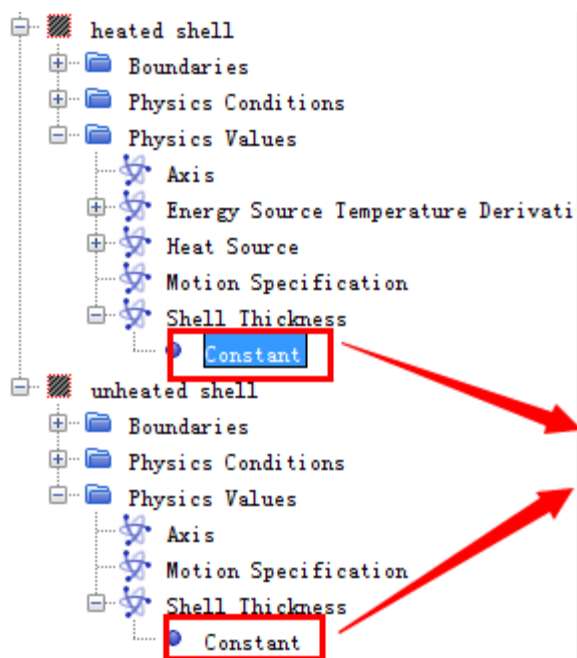
指定Shell Region的物理模型

1. 在[Regions]中选择[heated shell]以及[unheated shell]，在Properties属性窗口中指定其物理模型为Physics 2。



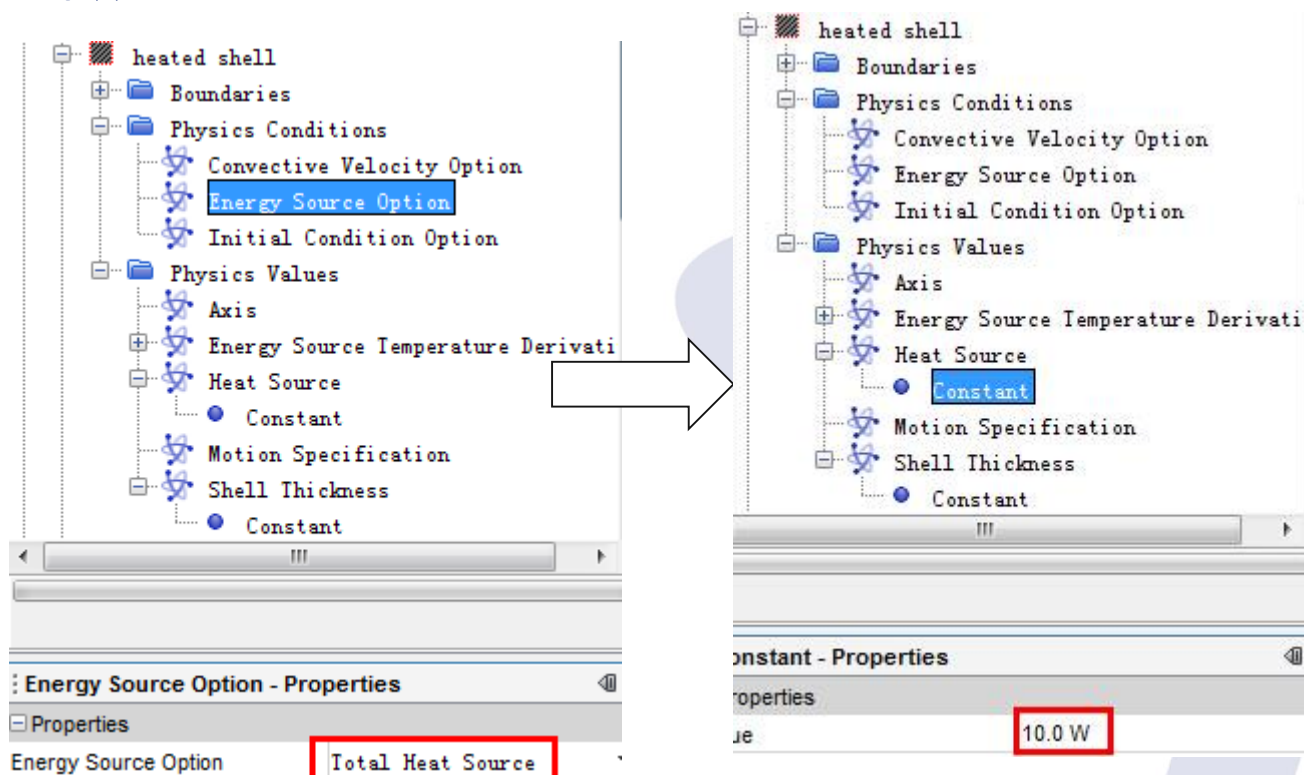
指定Shell的厚度

1. 选择[heated shell]和[unheated shell]，在[Physics Values] > [Shell Thickness]中输入 $1.0\text{e-}5\text{m}$ 。



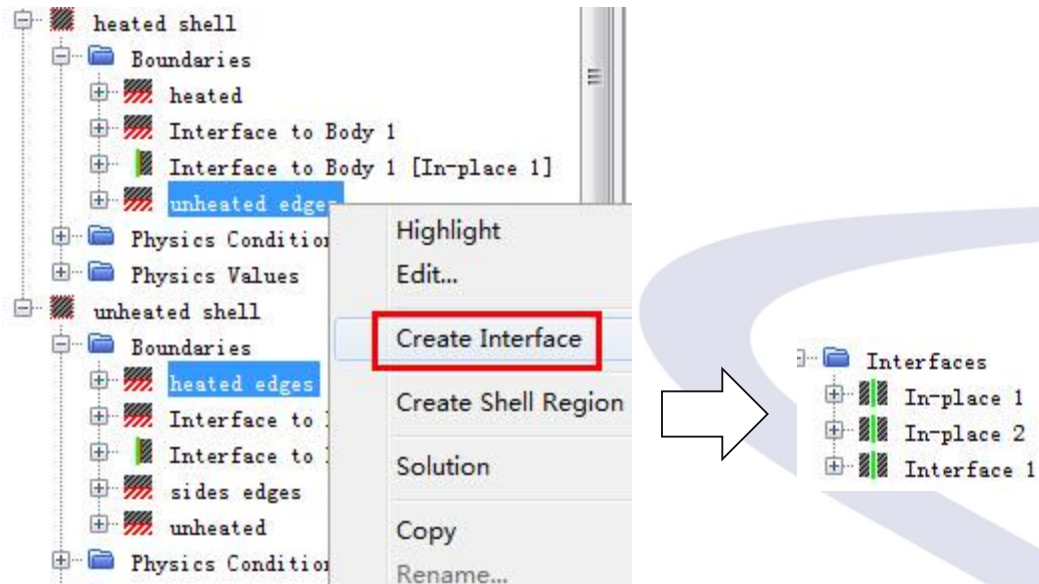
指定发热量

1. 在[heated shell] > [Physics Conditions] > [Energy Source Option]中，选择Total Heat Source。
2. 在[heated shell] > [Physics Value] > [Heat Source]中，设定其值为10W。



设定interface

1. 选择[heated shell]以及[unheated shell]、右击选择[Create Interface]。
 - 在Shell之间连接的Edge处生成interface。



实行计算和结果处理

1. 开始执行计算。

- 计算到一定程度可以终止计算。



2. 打开已有的标量云图。

- [Scalar Scene 1]

